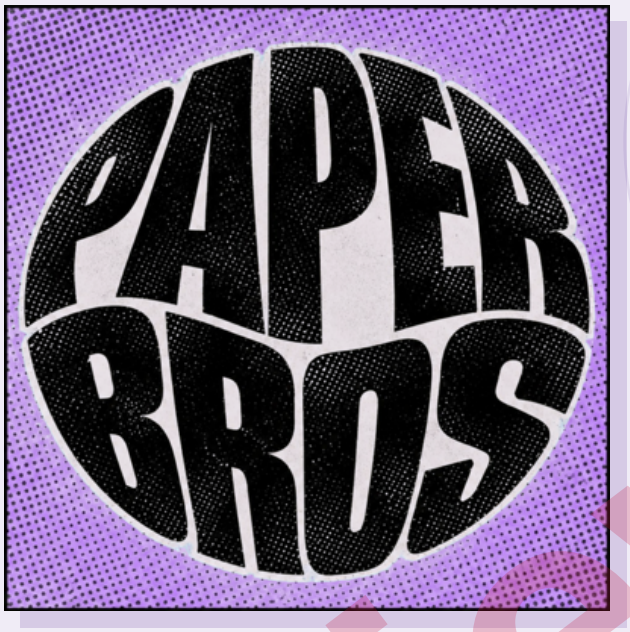
PREVIEW



CAMBRIDGE O LEVEL

FORMULA SHEET

Additional Mathematics 4037

Every formula you need for Paper 1 & Paper 2 · 2025–2027 syllabus

INSIDE

1 Functions
2 Quadratic functions
3 Factors of polynomials
4 Equations, inequalities & graphs
5 Simultaneous equations
6 Logarithms & exponentials
7 Straight-line graphs
8 Coordinate geometry of the circle
9 Circular measure
10 Trigonometry
11 Permutations & combinations
12 Series
13 Vectors in two dimensions
14 Differentiation
15 Integration & kinematics
16 Mensuration you need
17 Exam toolkit
18 Common mistakes
19 Key graph shapes

given = printed on the exam's List of formulas ▌Tip = exam technique

Preview

1 Functions

Composite $fg(x) = f(g(x))$ (do g first)

Inverse f^{-1}: write $y = f(x)$, make x the subject, then swap x and y.

Domain of f^{-1} = range of f

Range of f^{-1} = domain of f

Inverse property $ff^{-1}(x) = f^{-1}f(x) = x$

Notation $f^2(x) = f(f(x))$

For gf: domain of $gf \subseteq$ domain of f, and range of $gf \subseteq$ range of g.

The graph of $y = f^{-1}(x)$ is the reflection of $y = f(x)$ in the line $y = x$.

Modulus $$|x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

$y = |f(x)|$: reflect every part of $y = f(x)$ that lies below the x-axis in the x-axis.

> **Tip** Only a **one–one** function has an inverse. To make a quadratic one–one, restrict the domain to one side of its vertex, e.g. $x \geq -\frac{b}{2a}$. For gf to exist, the range of f must lie inside the domain of g.

2 Quadratic functions

Completing the square

$$ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 + c - \frac{b^2}{4a}$$

Vertex (turning point) $\left(-\frac{b}{2a},\ c - \frac{b^2}{4a}\right)$

$a > 0 \Rightarrow$ minimum point

$a < 0 \Rightarrow$ maximum point

Quadratic formula given $$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

DISCRIMINANT $b^2 - 4ac$

$b^2 - 4ac > 0$ two distinct real roots

$b^2 - 4ac = 0$ two equal (repeated) roots

$b^2 - 4ac < 0$ no real roots

$b^2 - 4ac \geq 0$ real roots

Line and curve: substitute the line into the curve to get one quadratic, then use its discriminant: > 0 meets at two points, $= 0$ line is a **tangent**, < 0 does not meet.

QUADRATIC INEQUALITIES ($\alpha < \beta$ ARE THE ROOTS, $a > 0$)

$(x - \alpha)(x - \beta) < 0$ $\alpha < x < \beta$

$(x - \alpha)(x - \beta) > 0$ $x < \alpha$ or $x > \beta$

> **Tip** Always sketch the parabola to read off an inequality. Write the answer as one interval ($-3 < x < 4$) or two separate ones ($x < 1$ or $x > 6$) — never "$1 > x > 6$".

3 Factors of polynomials

Remainder theorem $p(x) \div (x - a) \Rightarrow$ remainder $p(a)$

$p(x) \div (ax - b) \Rightarrow$ remainder $p(\frac{b}{a})$

Factor theorem $p(a) = 0 \iff (x - a)$ is a factor

Division identity $p(x) = (\text{divisor}) \times q(x) + \text{remainder}$

> **Tip** Try $x = \pm 1, \pm 2, \ldots$ (factors of the constant term) to find the first root; if the leading coefficient is not 1, also try $\pm\frac{p}{q}$ (p divides the constant, q the leading coefficient). Then find the quadratic factor by comparing coefficients or long division, and factorise or use the formula. To show $p(x) = 0$ has only one real root, show the quadratic factor has $b^2 - 4ac < 0$.

4 Equations, inequalities & graphs

$|ax + b| = c\ (c \geq 0)$ $ax + b = c$ or $ax + b = -c$

$|A| = |B|$ $A = B$ or $A = -B$

or square both sides: $A^2 = B^2$

$|A| < c$ $-c < A < c$

$|A| > c$ $A < -c$ or $A > c$

$|A| < |B|$ $A^2 < B^2$

For $|ax + b| = cx + d$, solve both cases and **reject** any answer that makes $cx + d < 0$.

SUBSTITUTION (HIDDEN QUADRATICS)

$x^{\frac{2}{3}} - 3x^{\frac{1}{3}} + 2 = 0$: let $u = x^{\frac{1}{3}}$. $3e^x = 12 - 5e^{-x}$: let $u = e^x$. $2(\ln 5x)^2 + \ln 5x - 6 = 0$: let $u = \ln 5x$. Solve for u, then undo; reject impossible values (e.g. $e^x = -2$, $\sqrt{x} = -3$).

CUBIC SKETCHES $y = k(x - a)(x - b)(x - c)$

- $k > 0$: rises from bottom-left to top-right; $k < 0$: the reverse.
- Crosses the x-axis at a, b, c; y-intercept: put $x = 0$.
- A squared factor $(x - a)^2$: the curve **touches** the axis at $x = a$.
- $y = |f(x)|$: reflect the parts below the axis; label all axis points.
- Cubic inequality $f(x) > d$: read off the x-values where the curve is above the line $y = d$.

5 Simultaneous equations

Make one variable the subject of the **linear** equation and substitute into the non-linear one. Solve the quadratic, then find the **pair** of values for each solution.

6 Logarithms & exponentials

Definition $y = a^x \iff x = \log_a y$

$(a > 0,\ a \neq 1,\ y > 0)$

Product $\log_a(xy) = \log_a x + \log_a y$

Quotient $\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$

Preview